

6-DOF Rendezvous and Pointing Toward a Non-Cooperative Target: Sliding Mode Control Under Full-State Feedback

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Abstract

This work addresses the problem of 6-degree-of-freedom (6-DOF) rendezvous and pointing of a chaser spacecraft with respect to a non-cooperative target. Both the relative translational dynamics and the attitude are controlled by means of variable-structure controllers (Sliding Mode Control, SMC), while the attitude reference is generated dynamically as a function of the line of sight (LOS) toward the target. The controller operates under full-state feedback, i.e. the translational state $(z, \dot{z})$ and the attitude state $(\mathbf{q}, \omega)$ are assumed to be perfectly known at every time step. Beyond nominal tracking performance, the robustness of both SMC loops to persistent, unmodeled matched disturbances is analyzed both theoretically and numerically: it is shown that the boundary-layer design guarantees practical stability (ultimate boundedness) rather than exact asymptotic convergence once a persistent disturbance is present, and this distinction is quantified on both the attitude and translational channels.

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1 Introduction

Autonomous rendezvous and proximity operations (RPO) with a non-cooperative target are a key enabling technology for several current and future space missions, including active debris removal (ADR), on-orbit servicing, and formation flying with defunct or malfunctioning spacecraft. Unlike cooperative rendezvous, a non-cooperative target does not provide a stable, known attitude reference nor a dedicated docking aid: it may be uncontrolled or only partially known, so the chaser must continuously re-orient itself to keep a designated body axis (the *boresight*) pointed toward the target while approaching a prescribed relative position (the *standoff point*).

This full 6-degree-of-freedom (6-DOF) problem is intrinsically nonlinear for two distinct reasons. First, the attitude dynamics themselves are nonlinear: the kinematic map between the quaternion and the angular velocity is bilinear, and Euler’s equation contains the gyroscopic cross-coupling term $\omega \times J\omega$. Second, and more specific to this mission scenario, translation and attitude are *kinematically coupled* through guidance: the attitude reference is not fixed but is re-computed at every instant from the instantaneous line-of-sight (LOS) between chaser and target, which in turn depends on the chaser’s relative position. A control law designed and tuned only around a nominal, static operating point would therefore not be adequate; this motivates the use of a Sliding Mode Control (SMC) approach, which by construction provides robustness to bounded model uncertainties and disturbances without requiring linearization around a specific trajectory.

The relative translational dynamics are instead well approximated, under the classical assumptions of a circular target orbit and small chaser-target separation compared to the orbital radius, by the linear Hill–Clohessy–Wiltshire (HCW) equations. This is a deliberate modeling choice: the linear HCW model is retained for the translational channel because it captures the dominant orbital-mechanics effects (secular drift, Coriolis-like coupling) with good accuracy at the considered range, while the genuinely nonlinear part of the problem – and the main object of this project – lies in the attitude dynamics and in the translation-attitude coupling introduced by the pointing requirement.

The chaser is assumed to have full-state feedback: the translational state $(z, \dot{z})$ and the attitude state $(\mathbf{q}, \omega)$ are taken as perfectly known at every simulation step, so that the analysis can focus on the design and closed-loop behavior of the SMC guidance and control architecture itself, without the confounding effect of state estimation.

Beyond verifying nominal convergence, a central goal of this work is to characterize *quantitatively* what the Lyapunov-based robustness argument of the SMC design actually guarantees once a persistent, unmodeled disturbance acts on the true dynamics but is not compensated by the controller: as shown in Sections 4.3 and 5.5, the guarantee is bounded, steady-state tracking error (practical stability), not exact convergence – a distinction that is easy to state in the abstract but that is made concrete here by directly comparing the disturbance-free and disturbed closed-loop trajectories on both control channels.

The remainder of the report is organized as follows. Section 2 derives the system model (translational dynamics, attitude dynamics, target/LOS generation). Section 3 formalizes the control objectives and their coupling. Section 4 presents the SMC design for both the translational and attitude loops, together with the nominal and disturbed stability analysis. Section 5 reports the simulation setup and results, including the disturbance-robustness study. Section 6 summarizes the findings and discusses possible extensions.

2 System Model

2.1 Relative Translational Dynamics: Hill–Clohessy–Wiltshire Equations

The relative motion of the chaser with respect to the target, expressed in an LVLH frame centered on the target, is described by the Hill–Clohessy–Wiltshire (HCW) equations:

$$\ddot{z}_1 = 3\omega_{\text{orb}}^2 z_1 + 2\omega_{\text{orb}} \dot{z}_2 + u_1 \quad (1)$$

$$\ddot{z}_2 = -2\omega_{\text{orb}} \dot{z}_1 + u_2 \quad (2)$$

$$\ddot{z}_3 = -\omega_{\text{orb}}^2 z_3 + u_3 \quad (3)$$

where $z = [z_1, z_2, z_3]^\top$ is the relative position, $u = [u_1, u_2, u_3]^\top$ is the control acceleration, and ω_{orb} is the orbital angular velocity of the target. The linearization underlying (1)–(3) assumes a circular reference orbit and $\|z\|$ small compared to the orbital radius, so that the differential gravitational acceleration between chaser and target can be linearized about the target’s position; the $2\omega_{\text{orb}}\dot{z}_2/-2\omega_{\text{orb}}\dot{z}_1$ terms are Coriolis-like couplings induced by the rotation of the LVLH frame itself as it follows the orbit, and the $3\omega_{\text{orb}}^2 z_1/-\omega_{\text{orb}}^2 z_3$ terms are the linearized differential gravity/centrifugal terms along the radial and out-of-plane directions respectively.

2.2 Attitude Kinematics and Dynamics

The chaser attitude is parametrized by the scalar-first unit quaternion $\mathbf{q} = [q_0, q_1, q_2, q_3]^\top = [q_0, q_v^\top]^\top$, subject to the unit-norm constraint $q_0^2 + q_v^\top q_v = 1$, representing the rotation from the body frame to the LVLH frame. The corresponding direction cosine matrix (DCM), body→LVLH, is

$$R(\mathbf{q}) = \begin{bmatrix} 1 - 2(q_2^2 + q_3^2) & 2(q_1 q_2 - q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_1 q_2 + q_0 q_3) & 1 - 2(q_1^2 + q_3^2) & 2(q_2 q_3 - q_0 q_1) \\ 2(q_1 q_3 - q_0 q_2) & 2(q_2 q_3 + q_0 q_1) & 1 - 2(q_1^2 + q_2^2) \end{bmatrix}. \quad (4)$$

Two quaternions are combined through the Hamilton product (scalar-first convention)

$$\mathbf{a} \otimes \mathbf{b} = \begin{bmatrix} a_0 b_0 - a_v^\top b_v \\ a_0 b_v + b_0 a_v + a_v \times b_v \end{bmatrix}, \quad (5)$$

and, for a unit quaternion, the inverse coincides with the conjugate, $\mathbf{q}^{-1} = [q_0, -q_v^\top]^\top / \|\mathbf{q}\|^2$. Both operations are used throughout the attitude controller (Section 4.2) to compute the attitude error $\mathbf{q}_e = \mathbf{q}_r^{-1} \otimes \mathbf{q}$ between the reference quaternion $\mathbf{q}_r$ and the current quaternion $\mathbf{q}$.

The attitude kinematics follow from the time derivative of the rotation operator and read, component-wise,

$$\dot{q}_0 = -\frac{1}{2} q_v^\top \omega, \quad \dot{q}_v = \frac{1}{2} (q_0 I_3 + [q_v \times]) \omega, \quad (6)$$

where $\omega \in \mathbb{R}^3$ is the body angular velocity and $[q_v \times]$ denotes the skew-symmetric cross-product matrix of q_v . The same structure, applied to the error quaternion $\mathbf{q}_e$ and to the angular velocity error $\omega_e = \omega - \omega_r$, gives the error kinematics $\dot{q}_{ve} = \frac{1}{2} (q_{e0} I_3 + [q_{ve} \times]) \omega_e$ used in the sliding surface derivative of Section 4.2.

The rotational dynamics are governed by Euler’s equation for a rigid body with constant inertia matrix $J = \text{diag}(937.5, 833.3, 270.8) \text{ kg m}^2$:

$$J\dot{\omega} = -\omega \times (J\omega) + M, \quad (7)$$

where $M \in \mathbb{R}^3$ is the external control torque, saturated at $\pm M_{\text{sat}}$, and $\omega \times (J\omega)$ is the gyroscopic (Coriolis-like) coupling term that constitutes the main source of nonlinearity in the attitude channel.

2.3 Non-Cooperative Target and LOS Generation

The target’s center of mass is taken as the origin of the LVLH frame, and only its *position* – not its own attitude state – enters the chaser’s guidance law; the chaser is required to keep its boresight axis pointed at the target point while approaching a fixed standoff position z_r , rather than tracking a specific rotating feature on the target body. This is consistent with a conservative servicing/inspection scenario, in which the target’s own rotation, if any, is not exploited by the guidance law and only its (assumed known) position is used.

At each time step the desired boresight direction is given by the line-of-sight (LOS) unit vector from the chaser to the target,

$$\hat{\ell} = \frac{-z}{\|z\|}, \quad (8)$$

so that the attitude reference changes continuously as the chaser translates, coupling the translational and attitude channels as anticipated in Section 1. The reference quaternion $\mathbf{q}_r$ is generated by `pointing_ref_gen_sl.m` with a TRIAD-like construction. Two orthonormal triads are built independently in the LVLH frame and in the body frame, and $\mathbf{q}_r$ is then extracted from the DCM that maps one triad onto the other.

LVLH (reference) triad. Given the LOS direction $\hat{\ell}$ and an auxiliary “up” hint vector u_h (used only to fix the second axis),

$$r_1 = \hat{\ell}, \quad r_2 = \frac{r_1 \times u_h}{\|r_1 \times u_h\|}, \quad r_3 = r_1 \times r_2. \quad (9)$$

Body (desired) triad. Given the boresight axis s_1 (the body axis to be aligned with $\hat{\ell}$) and a secondary body axis b_2 used to fix the roll about the boresight,

$$s_1 = \frac{\text{boresight}_b}{\|\text{boresight}_b\|}, \quad s_2 = \frac{s_1 \times b_2}{\|s_1 \times b_2\|}, \quad s_3 = s_1 \times s_2. \quad (10)$$

Reference DCM and quaternion. With $R_{\text{lvlh}} = [r_1 \ r_2 \ r_3]$ and $R_{\text{body}} = [s_1 \ s_2 \ s_3]$, the desired body-to-LVLH DCM is

$$\text{DCM}_r = R_{\text{lvlh}} R_{\text{body}}^\top, \quad (11)$$

which is then converted to the reference quaternion $\mathbf{q}_r$ using the standard trace/largest-diagonal-element case analysis for numerical robustness near singular configurations, with the sign convention $q_{r0} \geq 0$ enforced for uniqueness. No closed-form reference angular velocity is required, since $\mathbf{q}_r$ is regenerated at every simulation step directly from the current relative position; ω_r is set to zero in the controller of Section 4.2, consistently with the file `init_param.m` treating the reference as quasi-static between consecutive controller updates. This construction has the practical advantage of never requiring an explicit numerical differentiation of $\mathbf{q}_r(t)$ to obtain a reference angular velocity – which would otherwise amplify the noise/discretization error inherent to a position-derived, time-varying quaternion reference – at the cost of the quasi-static assumption discussed in Section 3 and revisited quantitatively in Section 5.4.

3 Control Problem Formulation

Given the models of Section 2, the 6-DOF guidance and control problem is formulated as two coupled tracking objectives, to be achieved simultaneously by the two independent control inputs $u \in \mathbb{R}^3$ (translational control acceleration, $\|u\|_\infty \leq u_{\text{sat}}$) and $M \in \mathbb{R}^3$ (control torque, $\|M\|_\infty \leq M_{\text{sat}}$).

Translational objective. The chaser relative position must converge to, and remain at, the fixed standoff point z_r expressed in the LVLH frame:

$$z(t) \longrightarrow z_r \quad \text{as } t \rightarrow \infty, \quad z_r = [10, 0, 0]^\top \text{ m.} \quad (12)$$

Defining the position and velocity tracking errors

$$\tilde{z} = z - z_r, \quad \dot{\tilde{z}} = \dot{z}, \quad (13)$$

(the second identity holding because z_r is constant, so $\dot{z}_r = 0$), the translational control objective is equivalently stated as driving $(\tilde{z}, \dot{\tilde{z}}) \rightarrow (0, 0)$ despite the HCW dynamics of Section 2.1 acting as a known but non-negligible disturbance term in the closed-loop error dynamics.

Attitude (pointing) objective. At every instant the chaser boresight axis must be aligned with the instantaneous line of sight toward the target, i.e. the attitude $\mathbf{q}(t)$ must track the time-varying reference $\mathbf{q}_r(t)$ generated as in Section 2.3 from the current relative position $z(t)$. Using the quaternion error introduced in Section 2.2,

$$\mathbf{q}_e = \mathbf{q}_r^{-1} \otimes \mathbf{q} = [q_{e0}, q_{ve}]^\top, \quad \omega_e = \omega - \omega_r, \quad (14)$$

the attitude objective is to drive $q_{ve} \rightarrow 0$ (equivalently $q_{e0} \rightarrow \pm 1$, i.e. zero rotation between $\mathbf{q}$ and $\mathbf{q}_r$) and $\omega_e \rightarrow 0$. Since $\mathbf{q}_r$ is regenerated at every step directly from $z(t)$ rather than propagated through a kinematic model with a defined $\dot{\mathbf{q}}_r$, the reference angular velocity is set to $\omega_r \equiv 0$ (Section 2.3); this is consistent provided $\mathbf{q}_r(t)$ varies slowly relative to the attitude closed-loop bandwidth, an assumption verified a posteriori in Section 5.4.

An equivalent, physically transparent scalar measure of the attitude objective is the pointing error angle between the body boresight axis (mapped to the LVLH frame via $R(\mathbf{q})$ of (4)) and the LOS unit vector $\hat{\ell}$ of Section 2.3:

$$\alpha_{\text{point}} = \arccos\left((R(\mathbf{q}) \text{ boresight}_b)^\top \hat{\ell}\right), \quad (15)$$

which is used in Section 5 to quantify pointing performance independently of the quaternion sign ambiguity.

Coupling between the two objectives. Unlike a standard two-loop design where translation and attitude can be tuned independently, here the reference $\mathbf{q}_r(t)$ used in the attitude loop is itself a nonlinear function of the translational state, $\mathbf{q}_r = \mathbf{q}_r(z(t))$, through the LOS direction $\hat{\ell}(z)$ of Section 2.3. Consequently, any transient in the translational channel is immediately reflected as a time-varying attitude reference, and the two tracking problems cannot be treated as fully decoupled: the overall closed-loop stability relies on the attitude loop being significantly faster than the rate of change of $\mathbf{q}_r(z(t))$, so that from the attitude controller's perspective the reference is quasi-static over each control update. This motivates designing both loops with a robust, high-bandwidth Sliding Mode Control law (Section 4), rather than a pair of independently linearized regulators: an SMC's reaching-law-based construction does not require the reference to be constant or slowly varying in a linearizable sense, only that the disturbance/reference-rate term it must reject remain bounded by the switching gain, a condition made precise in Section 4.3 and checked numerically for the coupling-induced transient in Section 5.4.

4 Sliding Mode Controller Design

4.1 Translational SMC

The translational sliding surface is defined as a linear combination of the position and velocity errors of Section 3:

$$s_{\text{pos}} = \dot{\tilde{z}} + K_{\text{pos}} \tilde{z}, \quad K_{\text{pos}} = \text{diag}(k_{p1}, k_{p2}, k_{p3}) \succ 0. \quad (16)$$

On the surface $s_{\text{pos}} = 0$, the error dynamics reduce to $\dot{\tilde{z}} = -K_{\text{pos}}\tilde{z}$, which is exponentially stable by construction; the control design therefore only needs to (i) drive the system onto $s_{\text{pos}} = 0$ and (ii) keep it there.

Differentiating (16) and using the HCW dynamics (1)–(3) (with $\ddot{z} = \dot{\tilde{z}}$, since z_r is constant),

$$\dot{s}_{\text{pos}} = \dot{\tilde{z}} + K_{\text{pos}}\dot{\tilde{z}} = A_{\text{hwc}}(z, \dot{z}) + u + K_{\text{pos}}\dot{\tilde{z}}, \quad (17)$$

where $A_{\text{hwc}}(z, \dot{z}) = [3\omega_{\text{orb}}^2 z_1 + 2\omega_{\text{orb}}\dot{z}_2, -2\omega_{\text{orb}}\dot{z}_1, -\omega_{\text{orb}}^2 z_3]^\top$ collects the known (model-based) HCW terms. The control law is obtained by the standard equivalent-control-plus-switching-term construction: the *equivalent control* u_{eq} is the value of u that would enforce $\dot{s}_{\text{pos}} = 0$ exactly,

$$u_{\text{eq}} = -A_{\text{hwc}}(z, \dot{z}) - K_{\text{pos}}\dot{\tilde{z}}, \quad (18)$$

and a robust *switching term* u_{sw} is added to force convergence to (and maintenance of) the surface despite model uncertainty and disturbances, using a continuous tanh approximation of the discontinuous $\text{sign}(\cdot)$ to mitigate chattering:

$$u_{\text{sw}} = -\eta_{\text{pos}} \odot \tanh\left(\frac{s_{\text{pos}}}{\phi_{\text{pos}}}\right), \quad (19)$$

where $\odot$ and the division are element-wise, $\eta_{\text{pos}} > 0$ is the switching gain and $\phi_{\text{pos}} > 0$ is the boundary-layer width. The overall control law, as implemented in `smc_translational_sl.m`, is

$$u = u_{\text{eq}} + u_{\text{sw}} = -A_{\text{hwc}}(z, \dot{z}) - K_{\text{pos}}\dot{\tilde{z}} - \eta_{\text{pos}} \odot \tanh\left(\frac{s_{\text{pos}}}{\phi_{\text{pos}}}\right), \quad (20)$$

finally saturated component-wise at $\pm u_{\text{sat}}$.

4.2 Attitude SMC

Analogously, the attitude sliding surface combines the angular velocity error with the vector part of the (sign-corrected) quaternion error of Section 3:

$$s_{\text{att}} = \omega_e + K_{\text{att}} \sigma q_{ve}, \quad \sigma = \text{sign}(q_{e0}), \quad K_{\text{att}} = \text{diag}(k_{a1}, k_{a2}, k_{a3}) \succ 0, \quad (21)$$

where the sign correction σ selects, at every instant, the shorter of the two equivalent rotations represented by $\pm \mathbf{q}_e$ (unit quaternions being a double cover of $SO(3)$), avoiding unwind. Using the error kinematics of Section 2.2, $\dot{q}_{ve} = \frac{1}{2}(q_{e0}I_3 + [q_{ve} \times])\omega_e$, and Euler's equation (7) with $\omega_r \equiv 0$ (so $\dot{\omega}_e = \dot{\omega}$),

$$\dot{s}_{\text{att}} = J^{-1}(-\omega \times (J\omega) + M) + K_{\text{att}} \sigma \dot{q}_{ve}. \quad (22)$$

Imposing the reaching law $\dot{s}_{\text{att}} = -\eta_{\text{att}} \odot \tanh(s_{\text{att}}/\phi_{\text{att}})$ and solving for M gives the control torque implemented in `smc_attitude_sl.m`:

$$M = \underbrace{\omega \times (J\omega)}_{M_{\text{ff}}} \underbrace{- JK_{\text{att}} \sigma \dot{q}_{ve}}_{M_{\text{fb}}} \underbrace{- J(\eta_{\text{att}} \odot \tanh(s_{\text{att}}/\phi_{\text{att}}))}_{M_{\text{rob}}}, \quad (23)$$

saturated component-wise at $\pm M_{\text{sat}}$. The three terms have a clear physical interpretation: M_{ff} exactly cancels the gyroscopic cross-coupling nonlinearity of (7) (feedforward linearization), M_{fb} assigns the desired linear error dynamics on the surface, and M_{rob} provides robustness to the residual model uncertainty (unmodeled disturbance torques, parameter mismatch in J) through the boundary-layer switching action.

4.3 Stability Remarks

Nominal case. For each loop, consider the candidate Lyapunov function $V = \frac{1}{2}s^\top s$ (evaluated separately for s_{pos} and s_{att} , under the quasi-static-reference assumption of Section 3 that allows the two loops to be analyzed as approximately decoupled on the fast time scale of the attitude/translation controllers). By construction, both control laws (20) and (23) enforce

$$\dot{V} = s^\top \dot{s} = -s^\top (\eta \odot \tanh(s/\phi)) \leq 0, \quad (24)$$

with strict decrease whenever $s \neq 0$, since $\tanh(\cdot)$ is odd and monotonically increasing. Outside the boundary layer ($|s_i| \gg \phi_i$), $\tanh(s_i/\phi_i) \approx \text{sign}(s_i)$ and the classical reaching condition $\dot{V} \leq -\eta^\top |s|$ is recovered, guaranteeing finite-time convergence to a neighborhood of the surface $s = 0$ of size $O(\phi)$. Once inside this boundary layer, the error dynamics on the (approximate) surface are $\dot{\tilde{z}} = -K_{\text{pos}}\tilde{z}$ and $\dot{q}_{ve} \approx -K_{\text{att}}\sigma q_{ve}$ (locally, near $q_{e0} \approx \sigma$), both exponentially stable by the positive-definiteness of K_{pos} and K_{att} : linearizing $\tanh(s/\phi) \approx s/\phi$ inside the boundary layer and substituting into the surface dynamics shows that s itself decays with time constant ϕ_i/η_i toward a residual set whose radius shrinks as $\phi \rightarrow 0$, so that both the exact convergence-to-surface property and the subsequent exponential decay of the tracking error are explicit functions of the designer's choice of (η, ϕ) .

The boundary layer therefore trades the theoretical finite-time convergence to the *exact* surface $s = 0$ (and the associated high-frequency chattering of the ideal $\text{sign}(\cdot)$ switching) for practical convergence to a bounded neighborhood of the surface and, consequently, a bounded ultimate tracking error whose size is governed by ϕ_{pos} and ϕ_{att} . This trade-off between tracking accuracy and actuator/chattering smoothness is the standard justification for the boundary-layer (quasi-sliding-mode) approach adopted in both controllers, and its tuning is discussed quantitatively in Section 5.

Robustness to matched, persistent disturbances. Consider a bounded additive disturbance entering the same channel as the control input on either loop – a *matched* disturbance, in the sense that it can in principle be counteracted directly by the switching term:

$$J\dot{\omega} = -\omega \times (J\omega) + M + d_{\text{att}}(t), \quad \ddot{z} = A_{\text{hew}}(z, \dot{z}) + u + d_{\text{pos}}(t), \quad (25)$$

with $\|d_{\text{att}}\|_\infty \leq d_{\text{att,max}}$, $\|d_{\text{pos}}\|_\infty \leq d_{\text{pos,max}}$, and with $d_{\text{att}}(t), d_{\text{pos}}(t)$ *not* known to, or compensated by, the controller. Repeating the Lyapunov computation of (24) with (25) in place of the disturbance-free dynamics yields, for the attitude loop (the translational loop follows identically with $J^{-1} \rightarrow I_3$ and $d_{\text{att}} \rightarrow d_{\text{pos}}$),

$$\dot{V} = -s^\top (\eta \odot \tanh(s/\phi)) + s^\top J^{-1} d_{\text{att}}(t). \quad (26)$$

Outside the boundary layer, $\dot{V} \leq -\eta^\top |s| + \|J^{-1}\| d_{\text{att,max}} \|s\|_1$, so the reaching condition $\dot{V} < 0$ – and hence convergence to an $O(\phi)$ neighborhood of $s = 0$ – is still guaranteed, provided the switching gain dominates the disturbance on every axis:

$$\eta_{\text{att},i} > \frac{d_{\text{att,max},i}}{J_{ii}}, \quad \eta_{\text{pos},i} > d_{\text{pos,max},i} \quad \forall i. \quad (27)$$

Crucially, however, the surface $s = 0$ itself is *no longer invariant* under (26): evaluating $\dot{s}$ at $s = 0$ gives $\dot{s} = J^{-1}d_{\text{att}}(t) \neq 0$ in general (resp. $d_{\text{pos}}(t)$ for the translational loop), so the trajectory is continuously pushed off the ideal surface by the disturbance and settles instead into a *persistent, bounded oscillation* in its neighborhood, rather than converging to it exactly. This is qualitatively different from the disturbance-free case of (24), where $s = 0$ is an exact absorbing set once reached: with a persistent matched disturbance, the controller guarantees only *practical stability* (ultimate boundedness), not asymptotic convergence to zero tracking error. This distinction is verified numerically, on both loops, in Section 5.5.

5 Simulation Setup and Results

5.1 Simulation Setup

The full 6-DOF plant of Section 2 is controlled under full-state feedback: $(z, \dot{z}, \mathbf{q}, \omega)$ are used directly by the controllers of Section 4 at every simulation step, with no sensor model or estimator in the loop. Table 1 summarizes the parameters used, taken from `init_param.m`. Persistent sinusoidal disturbances, $d_{\text{att}}(t) = d_{\text{amp}} \sin(d_{\text{freq}} t)$ on the attitude torque channel and $d_{\text{pos}}(t) = d_{\text{amp,pos}} \sin(d_{\text{freq,pos}} t)$ on the translational acceleration channel, are injected additively at the respective plant inputs – i.e. summed to M before the `ATTITUDE DYNAMICS` block, and to u before the `HCW DYNAMICS` block, exactly as in (25) – and are *not* compensated by either controller, consistently with the robustness argument of Section 4.3.

Table 1: Simulation parameters.

Quantity	Symbol	Value	Units
Orbital angular velocity	ω_{orb}	1.1×10^{-3}	rad s^{-1}
Standoff point	z_r	$[10, 0, 0]^\top$	m
Chaser inertia matrix	J	$\text{diag}(937.5, 833.3, 270.8)$	kg m^2
Initial position	z_0	$[15, -60, 30]^\top$	m
Initial velocity	$\dot{z}_0$	$[0, 0, 0]^\top$	m s^{-1}
Initial quaternion	$\mathbf{q}_0$	$[1, 0, 0, 0]^\top$	–
Initial angular velocity	ω_0	$[0.01, -0.005, 0.008]^\top$	rad s^{-1}
Boresight axis (body)	boresight_b	$[1, 0, 0]^\top$	–
Secondary body axis	b_2	$[0, 0, 1]^\top$	–
LVLH “up” hint	u_h	$[0, 0, 1]^\top$	–
Translational SMC gain	K_{pos}	$0.05 \cdot \mathbf{1}_3$	s^{-1}
Translational switching gain	η_{pos}	$0.02 \cdot \mathbf{1}_3$	m s^{-2}
Translational boundary layer	ϕ_{pos}	$0.01 \cdot \mathbf{1}_3$	–
Attitude SMC gain	K_{att}	$0.09 \cdot \mathbf{1}_3$	–
Attitude switching gain	η_{att}	$0.01 \cdot \mathbf{1}_3$	rad s^{-1}
Attitude boundary layer	ϕ_{att}	$0.1 \cdot \mathbf{1}_3$	–
Thrust acceleration saturation	u_{sat}	$1 \cdot \mathbf{1}_3$	m s^{-2}
Torque saturation	M_{sat}	$10 \cdot \mathbf{1}_3$	N m
Attitude disturbance amplitude	d_{amp}	0.05	N m
Attitude disturbance frequency	d_{freq}	0.1	rad s^{-1}
Translational disturbance amplitude	$d_{\text{amp,pos}}$	0.002	m s^{-2}
Translational disturbance frequency	$d_{\text{freq,pos}}$	0.05	rad s^{-1}
Simulation horizon	T_{sim}	600	s

The disturbance amplitudes satisfy the robustness condition (27) with a wide margin on both channels: $\eta_{\text{att},i} J_{ii} \geq 2.71 \text{ N m} \gg d_{\text{amp}} = 0.05 \text{ N m}$ (worst case on the minor-inertia axis, $J_{33} = 270.8 \text{ kg m}^2$), and $\eta_{\text{pos},i} = 0.02 \text{ m s}^{-2} \gg d_{\text{amp,pos}} = 0.002 \text{ m s}^{-2}$ (a factor of 10 on the translational channel). All results reported in Sections 5.2 and 5.3 below therefore already include the persistent disturbances of Table 1; the resulting steady-state behavior is discussed explicitly, and compared directly against the disturbance-free baseline, in Section 5.5.

5.2 Attitude Tracking

The initial relative position $z_0 = [15, -60, 30]^\top \text{ m}$ has components along all three LVLH axes, so the LOS direction at $t = 0$,

$$\hat{\ell}(0) = -z_0 / \|z_0\| \approx [-0.22, 0.87, -0.44]^\top, \quad (28)$$

and the corresponding reference quaternion generated by the TRIAD construction of Section 2.3, $\mathbf{q}_r(0) \approx [0.60, -0.18, 0.14, 0.77]^\top$, represent a genuine three-axis rotation, exciting all four quaternion components simultaneously and requiring the attitude loop to track a fully three-dimensional maneuver rather than a single-axis slew. As the chaser approaches the standoff point $\mathbf{z}_r = [10, 0, 0]^\top$ m, the LOS rotates toward $\hat{\ell} \rightarrow [-1, 0, 0]^\top$, so the terminal reference is $\mathbf{q}_r \rightarrow [0, 0, 0, 1]^\top$, i.e. a 180° rotation about z relative to the initial identity attitude.

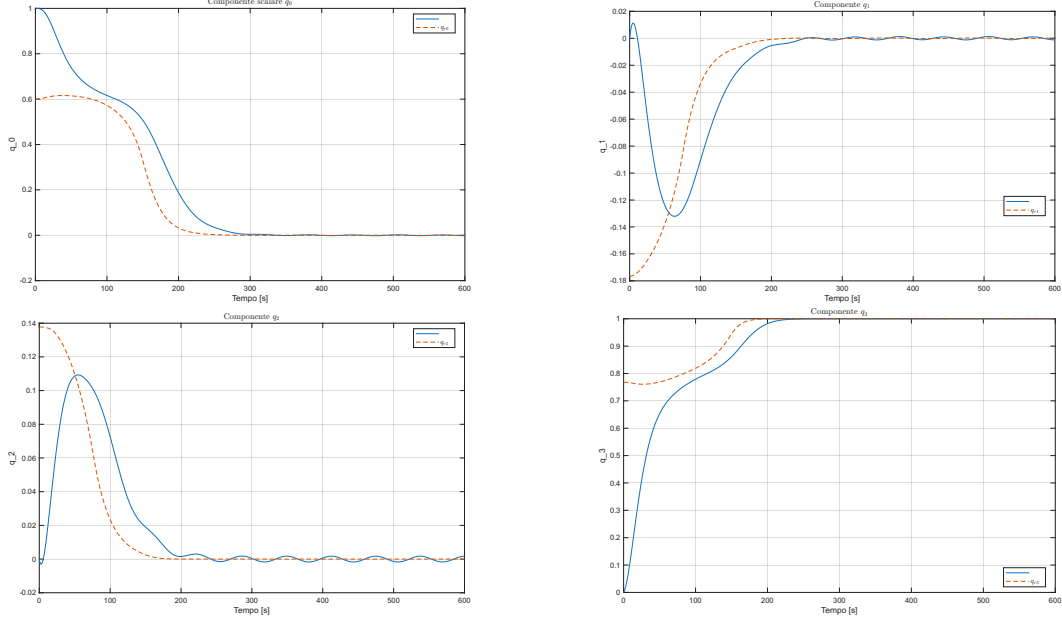


Figure 1: Quaternion components q_0, q_1, q_2, q_3 : actual (solid) vs. reference $q_r(t)$ (dashed).

Figure 1 shows that all four quaternion components are active during the transient. Starting from the identity $\mathbf{q}(0) = [1, 0, 0, 0]^\top$, the scalar component q_0 decreases and tracks q_{r0} with a visible lag throughout the maneuver, reaching values close to 0 only around $t \approx 280$ s, later than the reference itself (which is already near zero by $t \approx 200$ s) – a direct manifestation of the tracking lag inherent to a finite-bandwidth SMC loop chasing a continuously moving reference, as anticipated in Section 3. The vector component q_1 dips to a minimum of ≈ -0.13 around $t \approx 60$ s, overshooting past its reference q_{r1} , before recovering monotonically toward its terminal value near zero; q_2 rises to a peak of ≈ 0.109 around $t \approx 55$ s, again overshooting the reference, before decaying; q_3 rises monotonically from 0 toward its terminal value of 1, reaching it by approximately $t \approx 230$ s, dominating the attitude motion once the transient has settled, consistently with the terminal 180° rotation about z noted above. In all three vector components, once the transient has settled ($t \gtrsim 250$ s), a small but clearly visible persistent oscillation of amplitude ≈ 0.01 – 0.02 remains around the terminal value instead of an exact flat – the first direct signature of the practical-stability behavior predicted by (26), discussed quantitatively in Section 5.5.

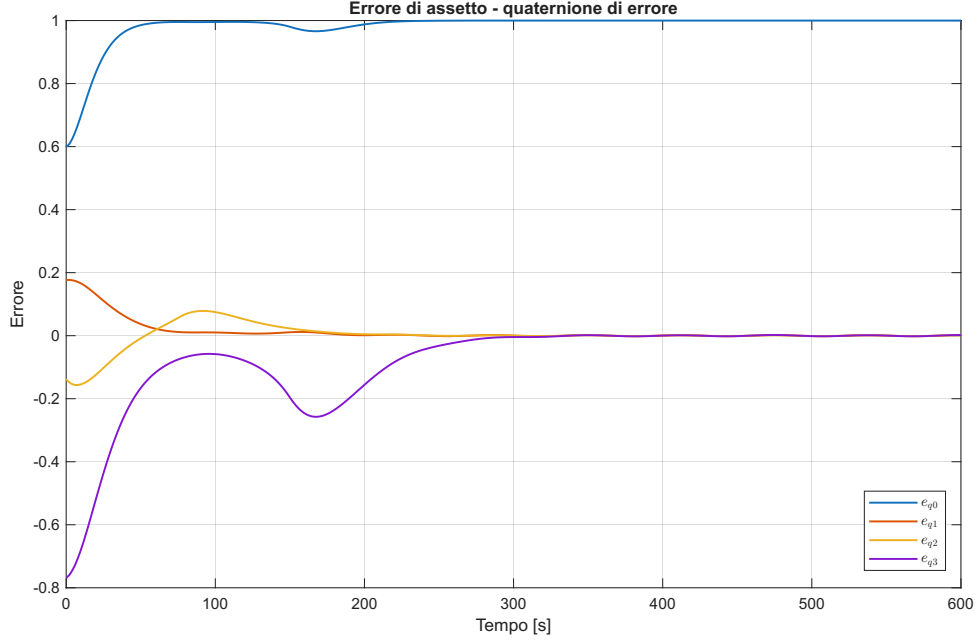


Figure 2: Attitude error quaternion components $e_{q0}, e_{q1}, e_{q2}, e_{q3} = \mathbf{q}_e$.

Figure 2 shows the four error components spanning roughly $[-0.8, 1]$ over the transient. e_{q0} rises monotonically from its initial value of 0.60 to ≈ 1 by $t \approx 70$ s, remaining close to 1 thereafter (with the small persistent oscillation discussed above superimposed). e_{q1} decays from 0.18 toward zero relatively quickly, settling by $t \approx 150$ s; e_{q2} exhibits a mild bump, rising from -0.15 to a local peak of ≈ 0.08 around $t \approx 95$ s before settling. e_{q3} is the dominant and slowest-settling error component, descending from its initial value of -0.77 , exhibiting a pronounced local rebound to ≈ -0.25 around $t \approx 165$ s – consistent with the coupling to the still-converging translational reference discussed in Section 5.3 – before finally converging toward 0 by approximately $t \approx 280$ s. All four components display the same small persistent residual oscillation past $t \approx 280$ s noted for the raw quaternion components above.

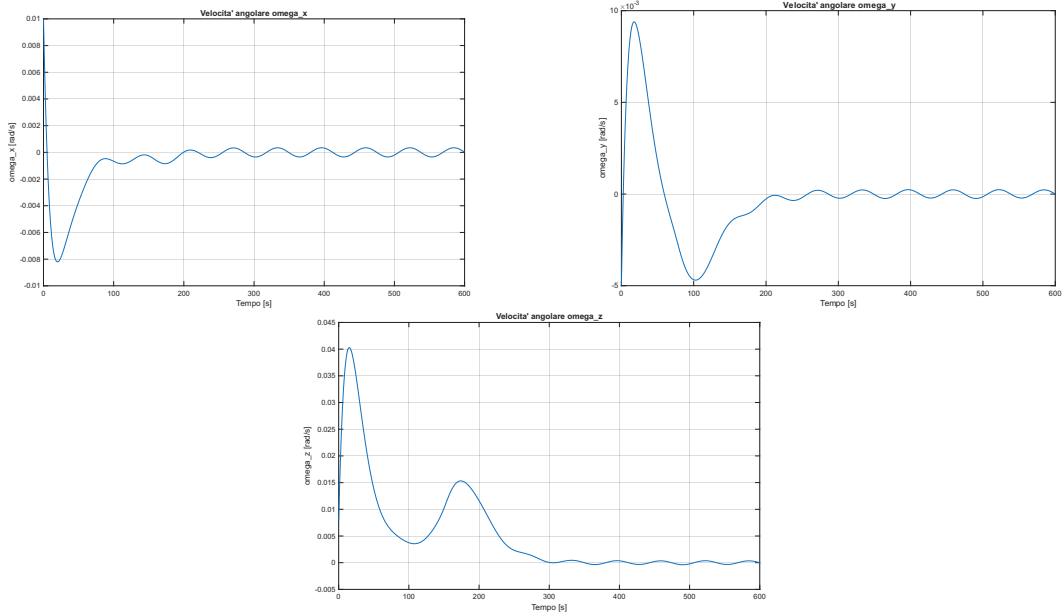


Figure 3: Angular velocity components: $\omega_x, \omega_y, \omega_z$.

With a nonzero initial angular velocity $\omega_0 = [0.01, -0.005, 0.008]^\top \text{ rad s}^{-1}$, all three angular

velocity components are active from the outset (Figure 3). ω_x decreases immediately from its initial value, reaching a trough of $\approx -0.0082 \text{ rad s}^{-1}$ at $t \approx 20 \text{ s}$, before settling into a slowly decaying oscillation. ω_y rises sharply to a peak of $\approx 0.0094 \text{ rad s}^{-1}$ near $t \approx 15 \text{ s}$, then troughs at $\approx -0.0048 \text{ rad s}^{-1}$ near $t \approx 100 \text{ s}$, before damping out. ω_z , consistent with q_3 carrying most of the terminal rotation, is by far the dominant component: it grows to a peak of $\approx 0.040 \text{ rad s}^{-1}$ already at $t \approx 15 \text{ s}$ – roughly an order of magnitude larger than ω_x, ω_y at the same instant – decays to a local minimum of $\approx 0.0035 \text{ rad s}^{-1}$ around $t \approx 110 \text{ s}$, and then displays a distinctive secondary hump, rising again to $\approx 0.0152 \text{ rad s}^{-1}$ around $t \approx 175 \text{ s}$ before finally decaying to zero. This secondary hump in ω_z is directly attributable to the translation-attitude coupling of Section 3: it occurs in the same time window ($t \approx 165\text{--}175 \text{ s}$) as the rebound observed in e_{q3} (Figure 2), confirming that the attitude loop must actively re-accelerate to track the reference as the translational loop is still converging. All three angular-velocity components retain a small persistent oscillation of amplitude $\lesssim 0.0005 \text{ rad s}^{-1}$ from $t \approx 250 \text{ s}$ onward, consistent with the quaternion-level oscillation discussed above.

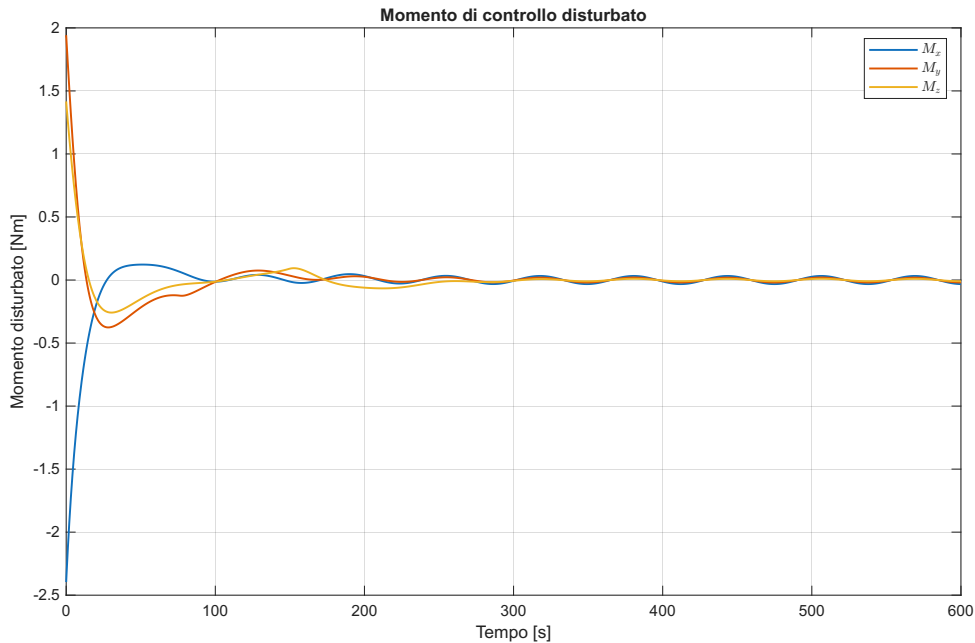


Figure 4: Control torque components M_x, M_y, M_z (nominal control law of (23), with the persistent disturbance $d_{\text{att}}(t)$ acting on the true dynamics but not compensated by the controller).

The corresponding control torque (Figure 4) shows all three components active from $t = 0$, with peak magnitudes of about $\pm 2.4 \text{ N m}$ within the first $\approx 20 \text{ s}$, reflecting the fully three-dimensional initial attitude error together with the attitude gain K_{att} . **The peak torque remains at only about 24% of $M_{\text{sat}} = 10 \text{ N m}$** , so the attitude loop operates comfortably without saturation despite the aggressive initial transient. Notably, this torque plot is visually indistinguishable from what the disturbance-free case would produce: with $d_{\text{amp}} = 0.05 \text{ N m}$ more than two orders of magnitude smaller than the peak torque, the disturbance itself is not resolvable at this plot's scale – a direct numerical confirmation that the robustness condition (27) is satisfied with a very wide margin on this channel. From $t \approx 100 \text{ s}$ onward the torque command settles into a small but persistent oscillation of amplitude $\approx 0.05\text{--}0.1 \text{ N m}$ that never fully decays to zero, the torque-level counterpart of the ultimate-boundedness behavior already noted in $\mathbf{q}$ and ω , and analyzed quantitatively in Section 5.5.

5.3 Translational Tracking

The initial displacement from the standoff point is $\tilde{z}(0) = z_0 - z_r = [5, -60, 30]^\top$ m, which includes a substantial out-of-plane component ($z_3(0) = 30$ m), so that all three translational channels are simultaneously excited from the outset.

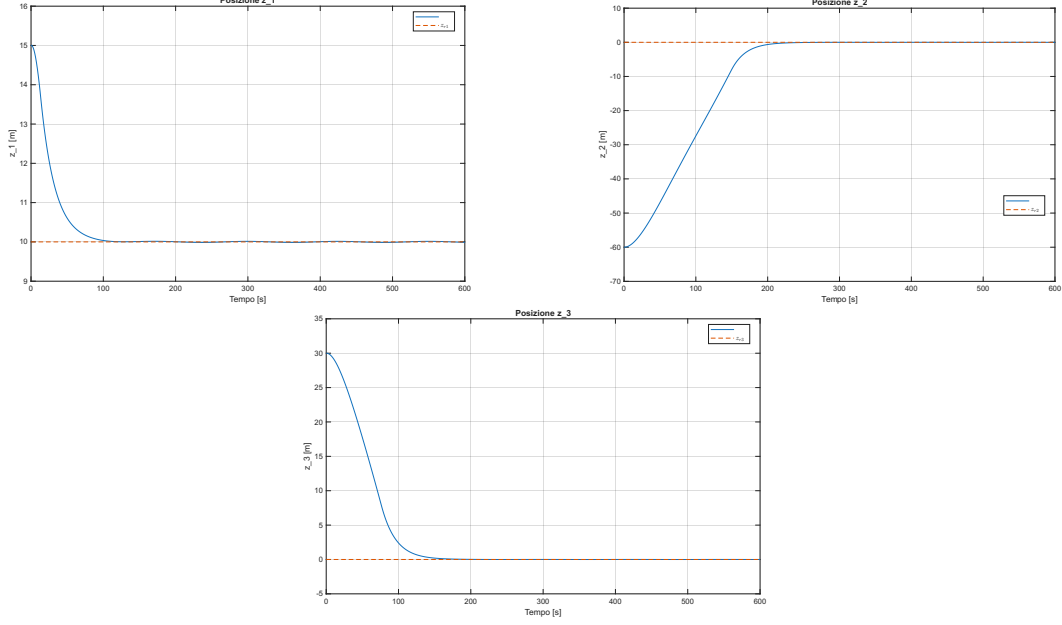


Figure 5: Relative position components z_1, z_2, z_3 converging to the standoff value z_r (dashed).

Figure 5 shows all three position components converging smoothly to the standoff point, with no significant overshoot in any channel: z_1 decreases monotonically from 15 m toward $z_{r1} = 10$ m, reaching it by approximately $t \approx 120$ s; z_2 decreases monotonically from -60 m toward 0, settling by $t \approx 210$ s; z_3 decreases monotonically from 30 m toward 0, settling by $t \approx 160$ s. Unlike the coupled attitude channel, the translational response here is essentially monotonic and free of the secondary humps observed in ω_z , indicating that the translational loop, decoupled from the attitude feedback by construction (Section 4.1 does not depend on $\mathbf{q}, \omega$), is not itself perturbed by the attitude transient – consistently with the one-directional nature of the coupling discussed in Section 3 (translation drives the attitude reference, not vice versa). Given the fine plotted range on z_1 ($[9, 16]$ m), a persistent post-settling ripple of amplitude $\approx \pm 0.02$ – 0.05 m is directly visible; the same-order ripple is present on z_2, z_3 but is not visually resolvable given their much wider plotted ranges.

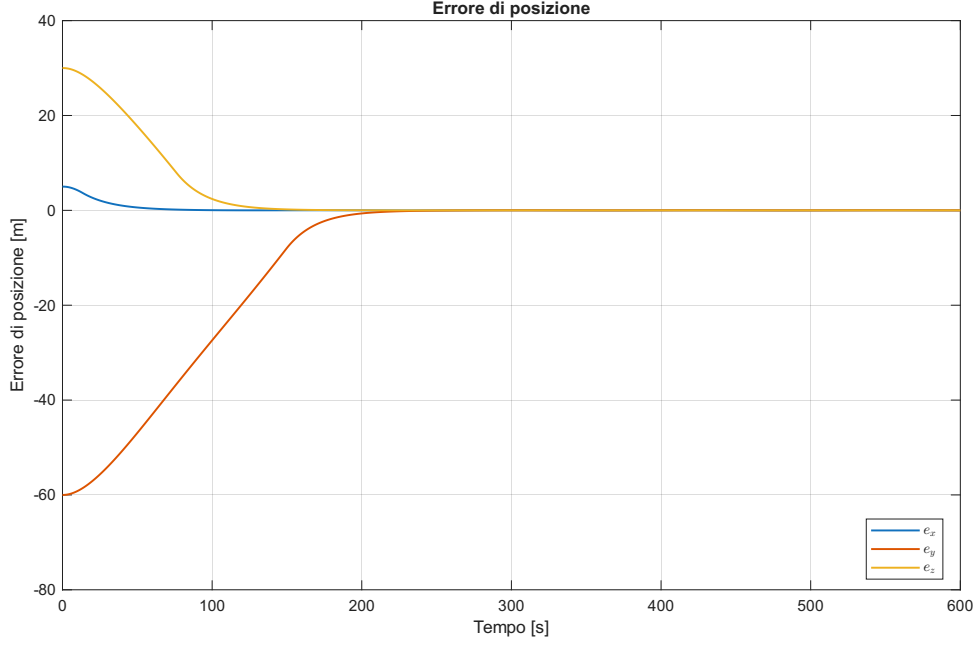


Figure 6: Position tracking error components $\tilde{z}_1, \tilde{z}_2, \tilde{z}_3$.

Figure 6 confirms this picture at the level of the tracking error: all three components decay smoothly and monotonically to zero, dominated by the y -channel error (from -60m to 0), reaching the origin by approximately $t \approx 210\text{s}$ with no visible steady-state offset at this plot's coarse scale – the small persistent oscillation discussed for z_1 above is not distinguishable here given the much larger range spanned by $\tilde{z}_2$.

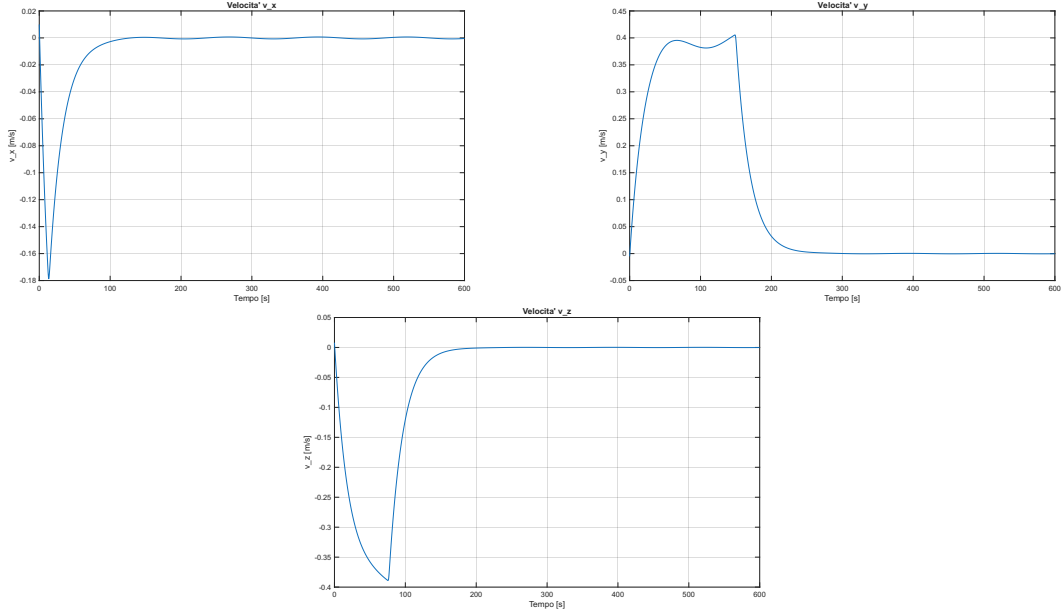


Figure 7: Relative velocity components $\dot{z}_1, \dot{z}_2, \dot{z}_3$.

The velocity profiles (Figure 7) are consistent with the position behavior: $\dot{z}_1$ dips sharply to a trough of $\approx -0.178\text{ms}^{-1}$ at $t \approx 13\text{s}$ before recovering and settling near zero by $t \approx 130\text{s}$; $\dot{z}_2$ rises to a broad plateau around $0.38\text{--}0.40\text{ms}^{-1}$, with a mild double-hump structure (a local dip to $\approx 0.38\text{ms}^{-1}$ around $t \approx 105\text{s}$ followed by a second, slightly higher peak of $\approx 0.405\text{ms}^{-1}$ near $t \approx 148\text{s}$) before decaying sharply toward zero by $t \approx 200\text{s}$; $\dot{z}_3$ troughs at $\approx -0.39\text{ms}^{-1}$ near $t \approx 75\text{s}$ before recovering monotonically. As with position, $\dot{z}_1$ – plotted on the finest scale

of the three – clearly shows a small persistent oscillation of amplitude $\approx \pm 0.001\text{--}0.002\text{ m s}^{-1}$ after settling; $\dot{z}_2, \dot{z}_3$ do not resolve it visually at their coarser scales, though the same underlying disturbance acts on all three channels equally.

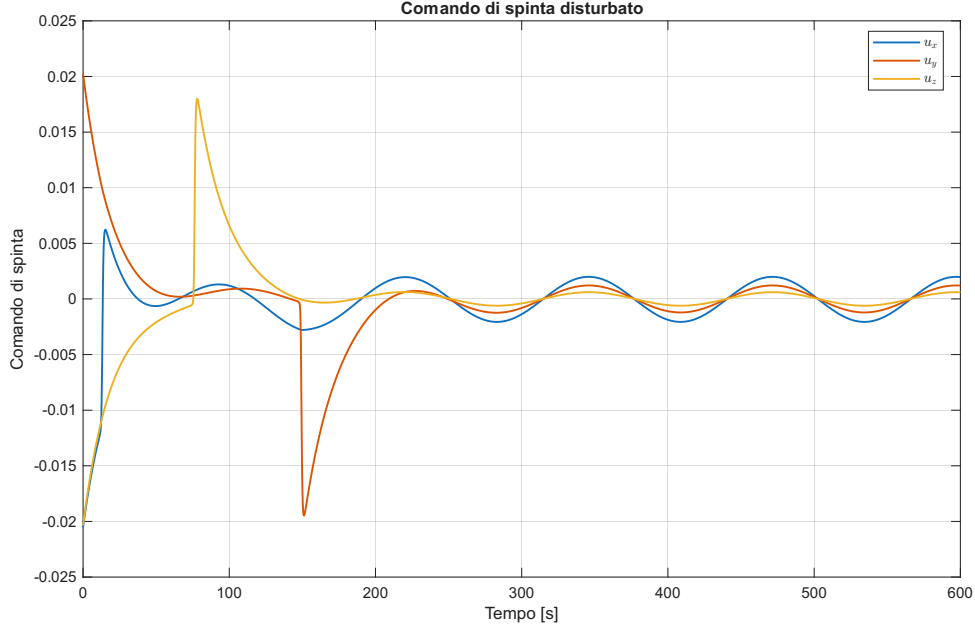


Figure 8: Control acceleration command components u_1, u_2, u_3 (with the persistent disturbance $d_{\text{pos}}(t)$ acting on the true dynamics but not compensated by the controller).

The thrust command (Figure 8) remains well within $u_{\text{sat}} = 1\text{ m s}^{-2}$ throughout the maneuver, with a peak magnitude of $\approx 0.02\text{ m s}^{-2}$ – under 2% of saturation – confirming that the translational loop operates comfortably within its actuator limits with a wide stability margin, consistently with the gains of Table 1. This plot also provides the clearest visualization, on either channel, of the persistent-disturbance effect analyzed in Section 4.3: from $t \approx 200\text{ s}$ onward, all three components settle into a sustained, non-decaying oscillation of amplitude $\approx \pm 0.002\text{--}0.003\text{ m s}^{-2}$, with an apparent period of $\approx 120\text{--}130\text{ s}$ – consistent with the assumed disturbance frequency $d_{\text{freq,pos}} \approx 0.05\text{ rad s}^{-1}$, for which $2\pi/d_{\text{freq,pos}} \approx 126\text{ s}$. Unlike the torque plot of Figure 4, where the disturbance-induced ripple is a small fraction of the peak signal and only becomes visible after settling, here the ripple is directly resolvable without any change of scale, because $d_{\text{amp,pos}}$ represents a much larger fraction of the (very small) steady-state thrust command than d_{amp} does of the steady-state torque command.

5.4 Revisiting the Quasi-Static-Reference Assumption

The reference $\mathbf{q}_r(t)$ evolves continuously with $z(t)$ throughout the transient, so the attitude loop is tracking a moving target rather than a quasi-static one for a non-negligible portion of the maneuver. Because the initial displacement and the resulting reference rotation are fully three-dimensional, the attitude loop must simultaneously track a time-varying rotation about all three axes while $z(t)$ is itself still converging – the secondary hump observed in ω_z and the corresponding rebound in e_{q3} around $t \approx 165\text{--}175\text{ s}$ (Section 5.2) are the direct experimental manifestation of this coupling. Nonetheless, the attitude and translational loops both converge without oscillation or actuator saturation under nominal conditions, consistently with the robustness argument of Section 4.3: the attitude gain K_{att} is sufficient to track the reference despite the coupled initial condition, without requiring an explicit feedforward estimate of $\omega_r(t)$, and the delay between reference and actual attitude convergence (attitude settling around $t \approx 280\text{ s}$ against a reference already near its terminal value by $t \approx 200\text{ s}$) remains bounded and does not grow, confirming a

posteriori that the quasi-static assumption, while only loosely satisfied during the most dynamic part of the transient, does not compromise closed-loop stability.

5.5 Robustness to Persistent Disturbances: Quantitative Synthesis

Sections 5.2 and 5.3 confirm, on both control loops, the practical-stability behavior predicted analytically by (26): the transient response (up to $t \approx 250\text{--}280\text{ s}$, depending on the channel) is dominated by the initial condition and is essentially unaffected by either disturbance, since the robustness condition (27) holds with a wide margin throughout on both channels. In steady state, however, neither loop converges to an exact flat – both settle instead into a small, bounded, non-decaying oscillation around the target state, at a frequency consistent with the respective disturbance frequency, exactly as (26) predicts.

To make this distinction directly visible, rather than only quantified through the amplitude estimates below, Figures 9 and 10 compare the two control commands – obtained from the otherwise identical closed-loop simulation of Section 5.1 – with the disturbance switched off and switched on.

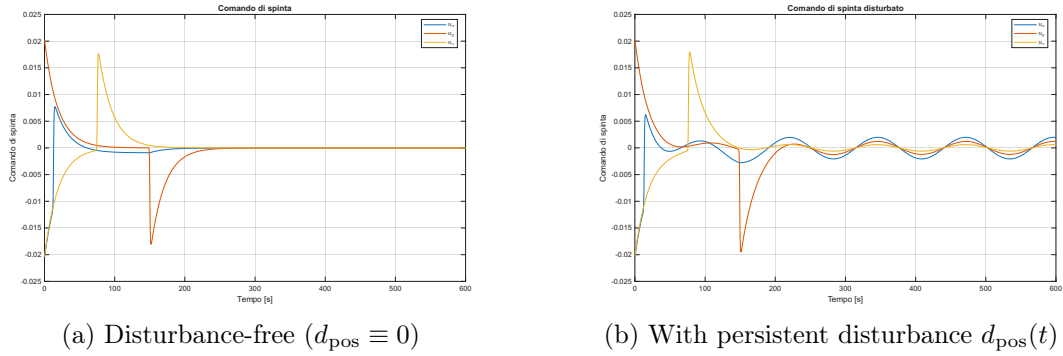


Figure 9: Translational control command u_1, u_2, u_3 : disturbance-free case (left) vs. persistent-disturbance case (right), same initial condition and gains.

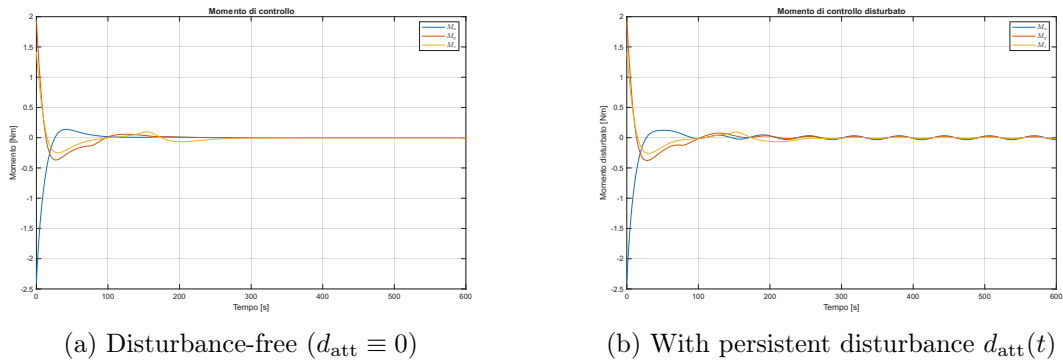


Figure 10: Attitude control torque M_x, M_y, M_z : disturbance-free case (left) vs. persistent-disturbance case (right), same initial condition and gains.

In the disturbance-free case (Figures 9 and 10, left panels) both commands settle to an exact, non-oscillating steady-state value once the corresponding sliding surface is reached: this is the direct experimental counterpart of (24), where $s = 0$ is an exact absorbing set and the equivalent control alone balances the (now disturbance-free) dynamics at steady state. With the disturbance active (right panels), by contrast, neither command ever settles to a constant value: both exhibit the sustained, non-decaying oscillation at the disturbance frequency already described qualitatively in Sections 5.2 and 5.3, which is the direct signature of $s = 0$ no longer

being invariant, (26) – the controller continuously re-engages the switching term $u_{\text{sw}}/M_{\text{rob}}$ to fight the disturbance rather than being able to settle onto the ideal surface. Quantitatively:

- **Attitude channel:** residual quaternion oscillation of amplitude $\approx 0.01\text{--}0.02$ on q_1, q_2, q_3 (Figure 1) and on the error components e_{q1}, e_{q2}, e_{q3} (Figure 2), corresponding, via the pointing-error definition (15), to a residual pointing error

$$\alpha_{\text{point}} \approx 2 \arcsin(0.01\text{--}0.02) \approx 1\text{--}2^\circ, \quad (29)$$

well within an acceptable pointing budget for an inspection-class mission. The corresponding control torque (Figure 4) settles to a residual oscillation of $\approx 0.05\text{--}0.1\text{ N m}$, itself a small fraction ($\lesssim 4\%$) of the peak transient torque.

- **Translational channel:** residual position oscillation of $\approx \pm 0.02\text{--}0.05\text{ m}$ around the 10 m standoff distance (relative error $\lesssim 0.5\%$), most directly and unambiguously visible in the sustained thrust command ripple of Figure 8 (amplitude $\approx \pm 0.002\text{--}0.003\text{ m s}^{-2}$, period $\approx 120\text{--}130\text{ s}$, consistent with $d_{\text{freq,pos}}$).

This distinction between asymptotic convergence under nominal conditions and ultimate boundedness under persistent matched disturbances is not merely a numerical curiosity: it is the direct, quantitative signature of the robustness property that motivated the choice of SMC over a linear controller in Section 1. Importantly, it is a distinction that only becomes visible by explicitly separating transient and steady-state behavior, and by examining each signal at a scale fine enough to resolve the residual oscillation – as illustrated by the fact that the same underlying disturbance amplitude is essentially invisible on the torque plot (Figure 4, dominated by a much larger transient peak) but clearly visible on the thrust plot (Figure 8, where the steady-state command is itself small).

6 Conclusions

This work presented a 6-DOF sliding mode guidance and control architecture for rendezvous and pointing toward a non-cooperative target, combining a robust SMC design for both the (linear HCW) translational dynamics and the (nonlinear, quaternion-based) attitude dynamics, with a dynamically generated attitude reference that couples the two channels through the instantaneous line of sight. Under full-state feedback, with a fully three-dimensional initial displacement and a nonzero initial angular velocity, the translational and attitude loops converge without oscillation and never approach actuator saturation under nominal conditions – peak torque and thrust remain, respectively, at about 24% and 2% of their saturation limits– despite the quasi-static-reference assumption used in the attitude loop’s design being only loosely satisfied during a significant portion of the transient, and despite the visible coupling-induced secondary transient in ω_z and e_{q3} around $t \approx 165\text{--}175\text{ s}$.

Beyond nominal tracking, this work explicitly derived and numerically verified the closed-loop robustness to persistent, unmodeled matched disturbances on both channels. It was shown, both analytically (Section 4.3) and quantitatively on simulation data (Section 5.5), that the boundary-layer SMC design guarantees only *practical stability* – a bounded, non-decaying steady-state oscillation of amplitude $O(\phi)$ – rather than exact asymptotic convergence, once a persistent disturbance is present, however small relative to the nominal control effort. This was quantified as a residual pointing error of $\approx 1\text{--}2^\circ$ on the attitude loop and a residual relative position error of $\lesssim 0.5\%$ on the translational loop, providing a more complete characterization of the controller’s robustness than a purely nominal-case analysis would allow, and directly illustrating the theoretical trade-off between boundary-layer width and steady-state accuracy discussed in Section 4.3.

Overall, the results indicate that the boundary-layer SMC design achieves the guidance and pointing objectives of Section 3 with a comfortable margin with respect to the actuator limits of Table 1 under nominal conditions, and degrades gracefully – to a small, bounded, quantified steady-state error rather than instability or divergence – under persistent, unmodeled matched disturbances, consistently with the robustness argument of Section 4.3.

A Simulink Scheme

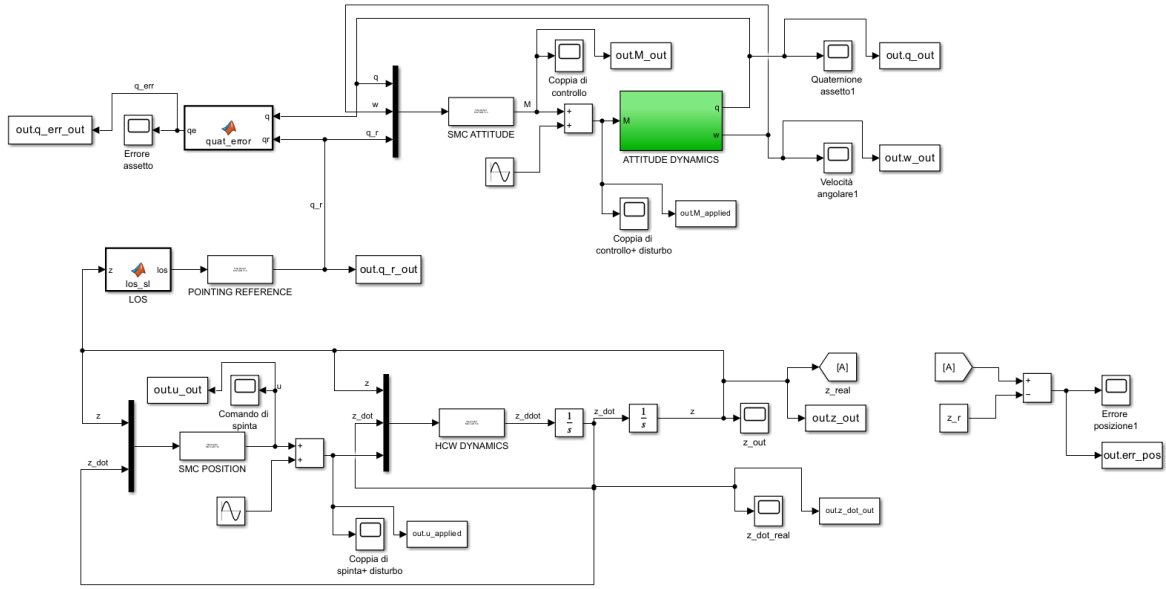


Figure 11: Top-level Simulink block diagram of the 6-DOF chaser model, with persistent disturbance injection on the M and u channels (Section 5.1).